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NEWS

New GitHub Copilot Research Finds 'Downward Pressure on Code Quality'

By David Ramel 01/25/2024

New research on the effect of AI-powered GitHub Copilot on software development cites some adverse results.

The "**Coding on Copilot**" whitepaper from GitClear sought to investigate the quality and maintainability of AI-assisted code compared to what would have been written by a human. In other words: "Is it more similar to the careful, refined contributions of a Senior Developer, or more akin to the disjointed work of a short-term contractor?"

The answer to that is summarized in this paragraph from the whitepaper's abstract:

"We find disconcerting trends for maintainability. Code churn -- the percentage of lines that are reverted or updated less than two weeks after being authored -- is projected to double in 2024 compared to its 2021, pre-AI baseline. We further find that the percentage of 'added code' and 'copy/pasted code' is increasing in proportion to 'updated,' 'deleted,' and 'moved' code. In this regard, AI-generated code resembles an itinerant contributor, prone to violate the DRY-ness [don't repeat yourself] of the repos visited."

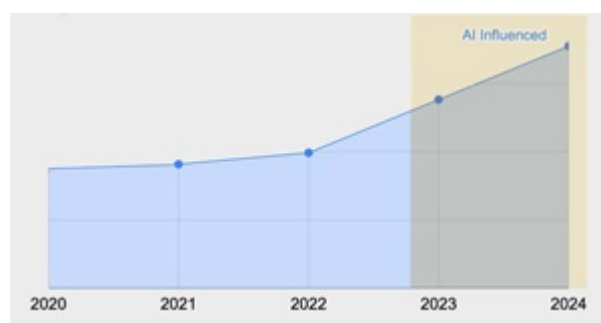
That serves as a counterpoint to findings of some other studies, including **one from GitHub in 2022** that found, for one thing: "developers who used GitHub Copilot completed the task significantly faster -- 55 percent faster than the developers who didn't use GitHub Copilot." That study was noted in the new whitepaper from GitClear, which sells a cloud-based code review tool. In addition to productivity, the GitHub study also measured positive effects in developer satisfaction and conserving mental energy.

GitClear's research, however, investigated "how the composition of code changes when AI is used." GitClear said its report sheds light on:

- What are the three significant changes since Copilot's introduction?
- What do Technical Leaders need to be on the lookout for 2024?
- How can you measure the impact of AI on your team's code quality?

Regarding that first item, the paper indicated the three most significant changes associated with Copilot's rise concerned "Churn" and "Moved" and "Copy/Pasted" code:

- **Burgeoning Churn:** "The bottom line is that 'using Copilot' is strongly correlated with 'mistake code' being pushed to the repo."



[Click on image for larger view.]

Code Churn by Year (source: GitClear).

- **Less Moved Code Implies Less Refactoring, Less Reuse:** "Combined with the growth in code labeled 'Copy/Pasted,' there is little room to doubt that the current implementation of AI Assistants discourages code reuse. Instead of refactoring and working to DRY ('Don't Repeat Yourself') code, these Assistants offer a one-keystroke temptation to repeat existing code."

- **More Copy/Pasted Code Implies Future Headaches:** "There is perhaps no greater scourge to long-term code maintainability than copy/pasted code. In effect, when a non-keyword line of code is repeated, the code author is admitting 'I didn't have the time to evaluate the previous implementation.' By re-adding code instead of reusing it, the chore is left to future maintainers to figure out how to consolidate parallel code paths that implement repeatedly-needed functionality."

The paper concludes: "How will Copilot transform what it means to be a developer? There's no question that, as AI has surged in popularity, we have entered an era where code lines are being added faster than ever before. The better question for 2024: who's on the hook to clean up the mess afterward?"

Some other studies on the topic include:

- **"Exploring the Verifiability of Code Generated by GitHub Copilot:** "We found evidence which corroborates the current consensus in the literature: Copilot is a powerful tool; however, it should not be 'flying the plane' by itself."
- **"Assessing the Quality of GitHub Copilot's Code Generation:** "Our empirical analysis shows that GitHub Copilot is a promising tool based on the results we obtained, however further and more comprehensive assessment is needed in the future."
- **Sea Change in Software Development: Economic and Productivity Analysis of the AI-Powered Developer Lifecycle:** "As more developers embrace these tools and acquire proficiency in the art of prompting with generative AI, it becomes evident that this novel approach to software development has forged a unique inextricable link between humans and artificial intelligence. This symbiotic relationship has the potential to shape the construction of the world's software for future generations."
- **The Impact of AI on Developer Productivity: Evidence from GitHub Copilot:** "Observed heterogenous effects show promise for AI pair programmers to help people transition into software development careers."
- **Study of software developers' experience using the Github Copilot Tool in the software development process:** "The results of the research indicate that developers' opinions are divided. Most of them met Github Copilot before attending the survey. The attitude to the tool was mostly positive but not many participants were willing to use it. Concerns are caused by security issues associated with using of Github Copilot."

For its study, GitClear collected and analyzed 153 million changed lines of code, authored between January 2020 and December 2023.

About the Author

David Ramel is an editor and writer at Converge 360.

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N

NCR

a year ago

AI does not holistically look at a project and try to understand it first. It's just sophisticated pattern matching, looks good on the surface, but creates maintenance headaches later.

14 2 Reply

**Fuck Trump**[→ NCR](#)

a year ago

It does if you fine tune it on your code base.

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**osteel**

a year ago

"or more akin to the disjointed work of a short-term contractor" that's a bit gratuitous

1 0 Reply

N

Natasha Upchurch

a year ago

Dry is a tertiary pattern anyways. Llms work better when the whole context of what you're working on is provided to them within 1k tokens.

2 2 Reply

G

Glenn Packer

a year ago edited

I think you should rephrase your words 'disjointed work of a short-term contractor?' that is very inflammatory and very insulting to many contractors.

Personally I don't really use it and i'm on a project with a lot of bad code and it is trying to put more very similar tech debt in.

I can see some useful bits especially around unit test setup but like all tools you need to be careful on proper use.

1 1 Reply **PB**

a year ago

Copilot is good at guesing a single line of code, sometimes a few more. But it isnt on par with these days AI's despite people pay for it. Not sure why MS is missing this chance, agents who talk about about your documents of code its so easy to do these days.. but no instead a braindead copilot..

1 1 Reply **Fuck Trump** PB

a year ago

It's far better than that.

1 0 Reply **J****Jesse Rudolph**

a year ago

It would have been more informative to see a higher resolution graph. There are quite a lot of relatively recent factors other than copilot that could explain this.

0 0 Reply 

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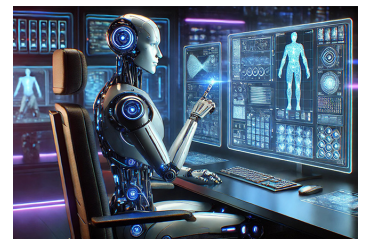
Visual Studio Devs Share Copilot AI Prompts to Improve Code

Microsoft's Mads Kristensen took to social media to ask Visual Studio developers to share their favorite prompts to get GitHub Copilot AI to improve their code.



Azure AI Foundry Gets 'Computer-Using Agent' for Autonomous GUI Interaction

Microsoft is expanding functionality for agentic AI into its Azure AI Foundry platform, furthering one of the hottest areas of development right now where AI controls computers just like humans.



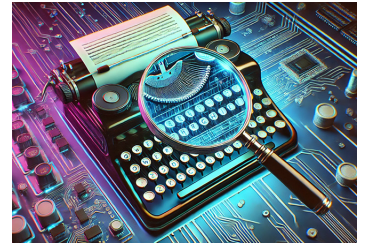
Microsoft Previews GPT-4o Copilot Code Completion and .NET AI Template in Visual Studio



Things are happening quickly in the Microsoft-centric AI dev space, with the company previewing new AI features in Visual Studio 2022 ranging from a new .NET AI template to GPT-4o Copilot Code Completion.

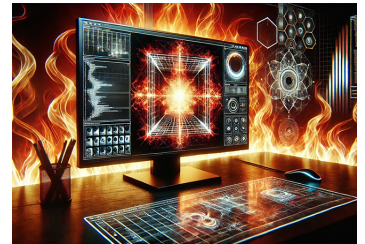
Microsoft Ports TypeScript to Go for 10x Native Performance Gains

Microsoft is revamping its TypeScript programming language with a native compiler and toolset. This effort seeks to address performance challenges, especially in large codebases, by porting the existing TypeScript compiler from TypeScript/JavaScript to the native language, Go.



Uno Platform Takes 'Hot Design' for Cross-Platform .NET Apps to Public Beta

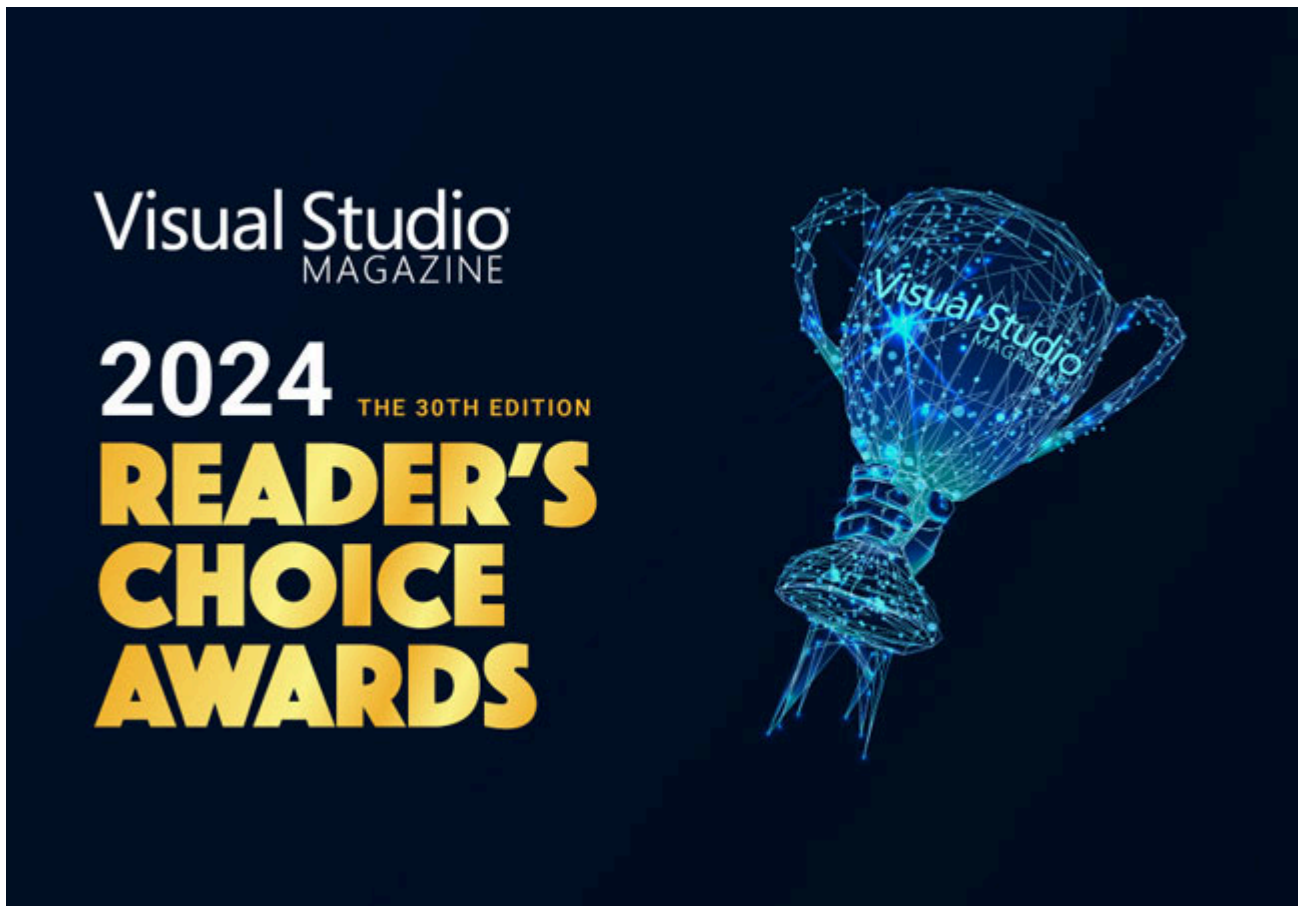
Uno Platform has taken its "Hot Design" feature to public beta, extending the "Hot Reload" paradigm with functionality to visually edit and refine a running app in real-time.



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